Electrical Machines (EE 2005)

Prof. Kamalesh Hatua

3+(1) classes/week — Tutorial

Quiz1 (20)
Quiz2 (20)
End sem (60)
} Tentative

Lab session (odd, Even)

Transformer
DC Machine
Induction machine
synchronous machine

Monday; 8:00 - 9:00
Tuesday; 1:00 pm - 2:00 pm
Thursday; 11:00 - 12:00
Friday; 10:00 - 11:00

Tutorial

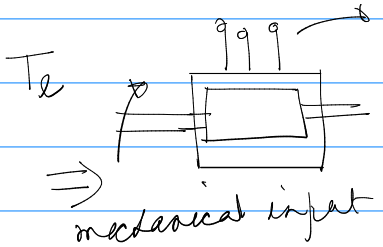
6 set → Please solve the problems independently

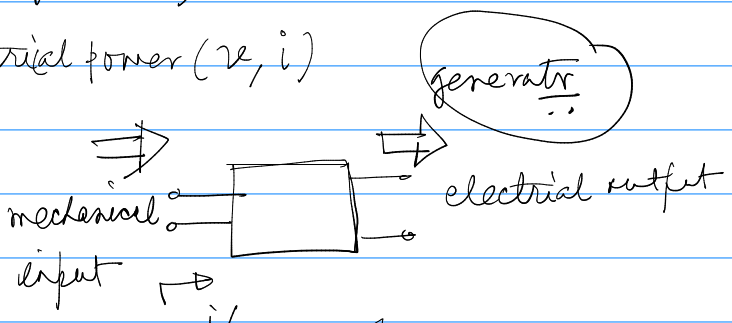
Electrical Machines

Fan (Motor)

Rotating machines (Motor, generator) — Static Machines (Transformer)

electrical power (v, i)





D.C motor — generator

D.C supply → D.C motor

Targeted efficiency (η) → 100%.

5%.

mechanical output — motor (Fan) — electrical input

A.C supply → 1) Induction motor
2) synchronous motor / n generator

95%.

Static machine (Transformer)

↳ steps up / steps down A.C voltage

11 kV 3ϕ 440 — 230 V

generator = 1 kV 3ϕ 400 kV — 3ϕ 230 V

1) Faraday's law of induction

2) KCL, KVL electrical circuit

3) Ampere's law

$$e = -n\frac{d\phi}{dt}$$

magnetic circuit

1φ AC

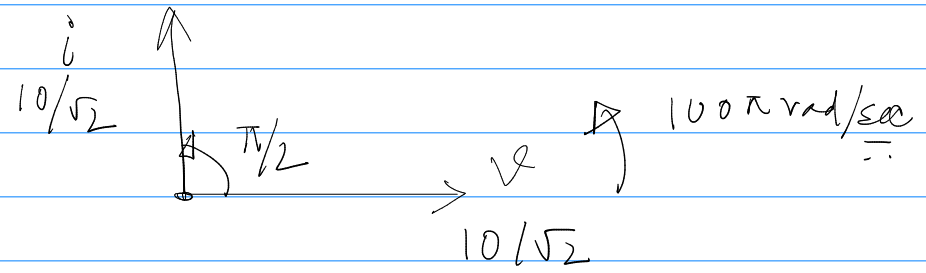
3φ AC → (To be done)

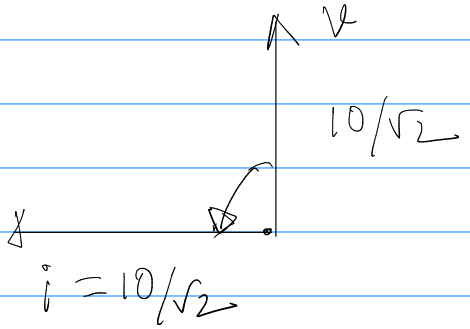
100π rad/sec

$v = 10 \sin 2\pi \cdot 50t = 10 \sin \omega t$

$i = 10 \sin(2\pi 50t + \pi/2)$

$(10/\sqrt{2})$





$v = 10 \sin 2\pi \cdot 50t$

$i = 10 \sin(2\pi 100t)$

power factor

$v = 10 \sin 2\pi 50t$

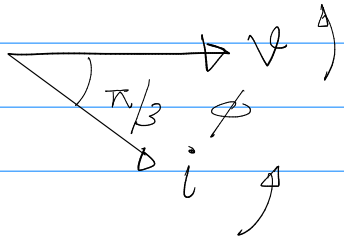
$i = 10 \sin(2\pi 50t - \pi/6)$

p.f. $\cos \pi/6$

lead

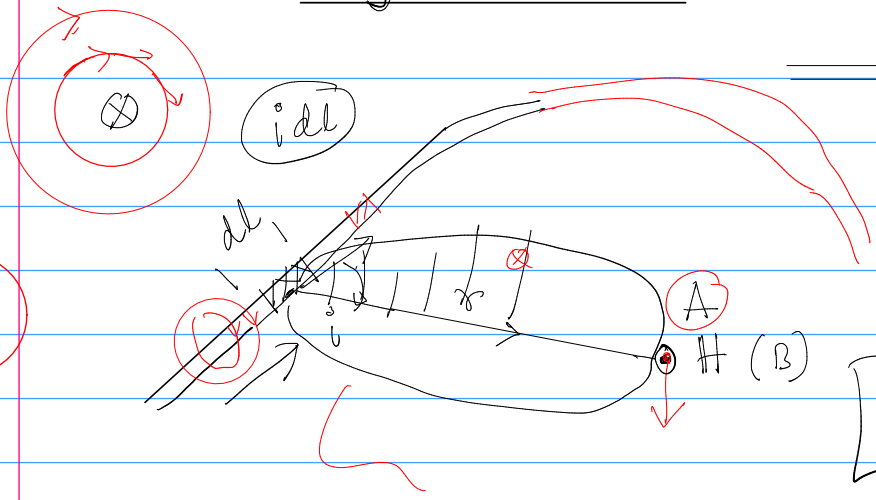
$\cos\phi = \cos(\pi/6)$

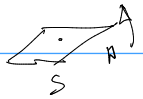
$= \sqrt{3}/2$ lag



Magnetic Circuit

20/01/2026





Biot savart's Law

$$\vec{H} \propto \frac{(i\,dl)}{r^2}$$

$$H = \frac{i\,dl \times \hat{a}_r}{r^2}$$

point from coil

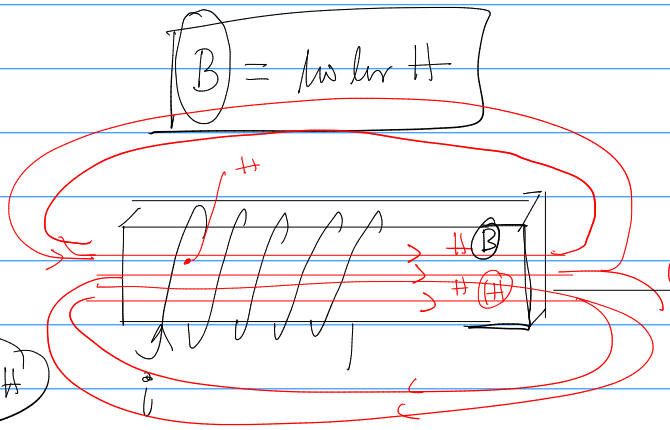
$B = \mu_0 \mu_r H$

$\mu_0 = 4\pi \times 10^{-7}$

$\mu_r = 3000$ (iron)

μ_r air = 1

$B = \mu_0 \mu_r H$



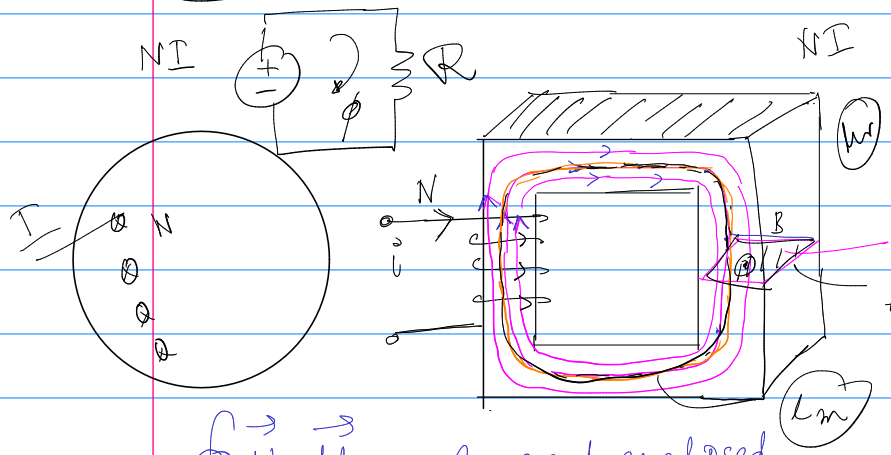
H

iron

solenoid

B = high iron

B = low air



H field

B

l_m = mean length of the magnetic ckt

A_c

Amperes law

$$\oint \vec{H} \cdot \vec{dl} = \text{current enclosed}$$

H, dl

$$\oint H \cdot dl = H \cdot l_m = NI$$

$$H = \frac{NI}{l_m}$$

$$NI = H\,l_m = \text{MMF} = \text{Magnetomotive force}$$

$B = \mu_0 \mu_r H$

$$NI = \frac{B}{\mu_0 \mu_r} l_m$$

$$B = \frac{\phi}{A_c}$$

$$NI = \frac{\phi\, l_m}{\mu_0 \mu_r A_c}$$

$$NI = \left(\frac{l_m}{\mu_0 \mu_r A_c}\right) \cdot \phi$$

$$NI = \mathcal{R} \cdot \phi$$

$\mathcal{R}$ = reluctance $= \left(\frac{l_m}{\mu_0 \mu_r A_c}\right)$

MMF = Reluctance * (flux)

electrical ckt magnetic ckt

emf

$V = Ri$

$V \Rightarrow NI$ (MMF)

$R \Rightarrow \mathcal{R}$ (reluctance)

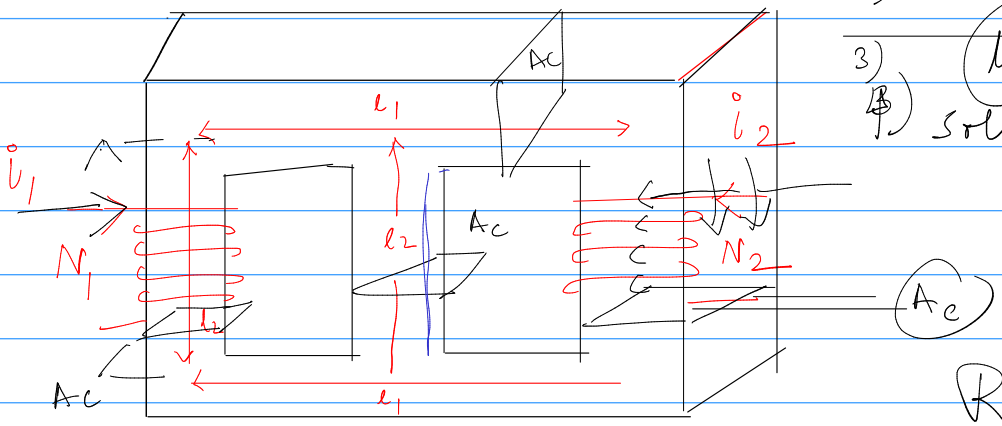
$i \Rightarrow \phi$ (flux)

$R = \frac{\rho l}{A}$

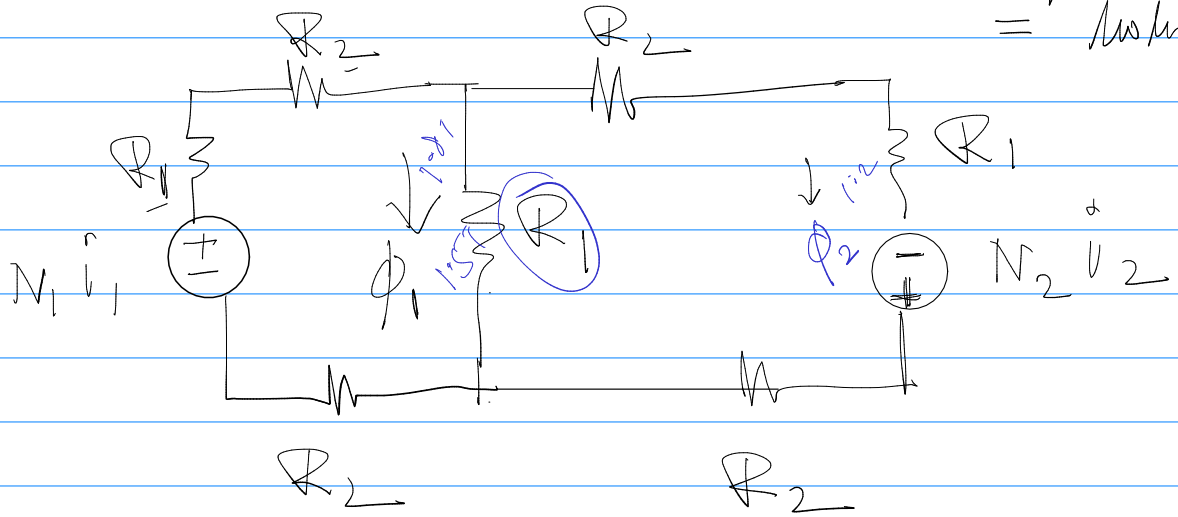
$\mathcal{R} = \frac{l_m}{\mu_0 \mu_r \cdot A_c}$



1) Identify the sources
2) Identify the reluctance path
3) Draw the equivalent magnetic ckt
4) Solve for the ckt



$\mathcal{R}_1 = \frac{l_2}{\mu_0 \mu_r A_c}$



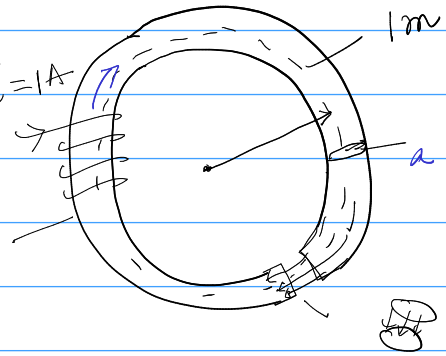
$\mathcal{R}_2 = \left(\frac{l_1}{\mu_0 \mu_r 2 A_c} \right)$

Class-3

Magnetic Circuit

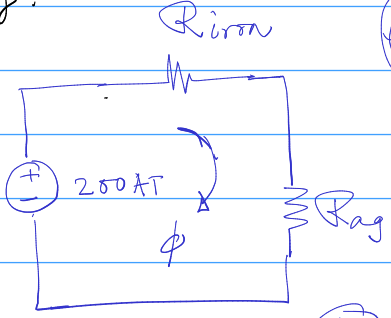
22/01/2026

Q) An iron ring of mean length 1 m has an air gap of 1 mm and a winding of 200 turns. If the relative permeability of iron is 500 when a current of 1A flows through the coil, find the flux density?



$R_{iron} = \frac{l}{a \mu_0 \mu_r}$

$R_{ag} = \frac{l_{ag}\ (1mm)}{a \mu_0}$



B = T, Wb/m²

B =

$NI = 200 = \phi \times [R_{iron} + R_{ag}] = \phi \left[\frac{l_{iron}}{a \cdot \mu_0 \mu_r} + \frac{l_{ag}}{a \cdot \mu_0}\right]$

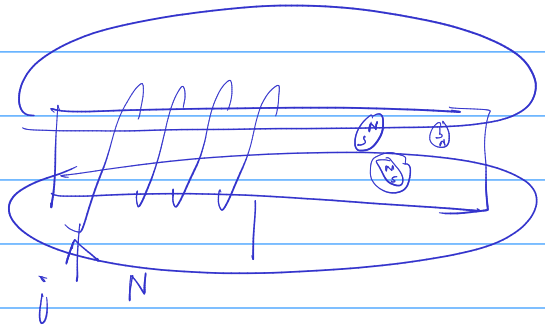
$200 = \frac{\phi}{a}\left[\frac{l_{iron}}{\mu_0 \mu_r} + \frac{l_{ag}}{\mu_0}\right]$ $B = 0.0838\ Wb/m^2$

200 = B [ ]

$\mu_r = B$

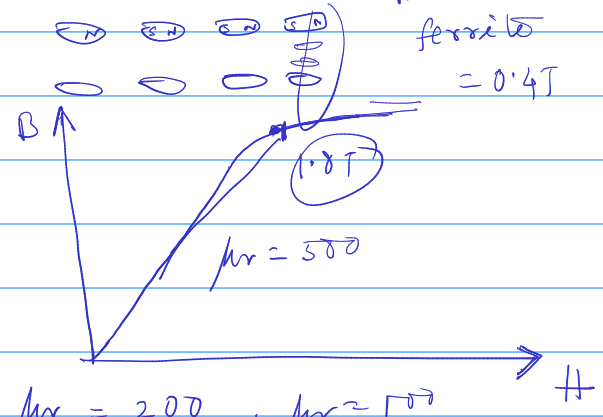
I = 10A ⇒ B = 0.838 Wb/m²

I = 100A ⇒ B = 8.38 Wb/m²

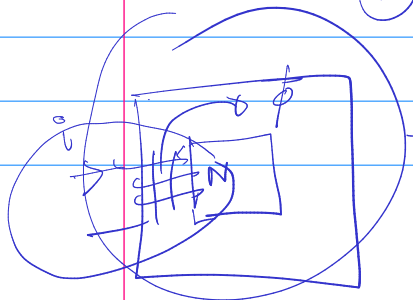


H = Ni

$\mu_r = \frac{B}{\mu_0 H}$



μr = 200, μr = 500



$\phi = \frac{NI}{R}$

L = Inductance

$L = \frac{\text{Total flux linkage}}{\text{current}}$

$L = \frac{N\phi}{I} = \frac{N(NI)}{I \cdot R}$

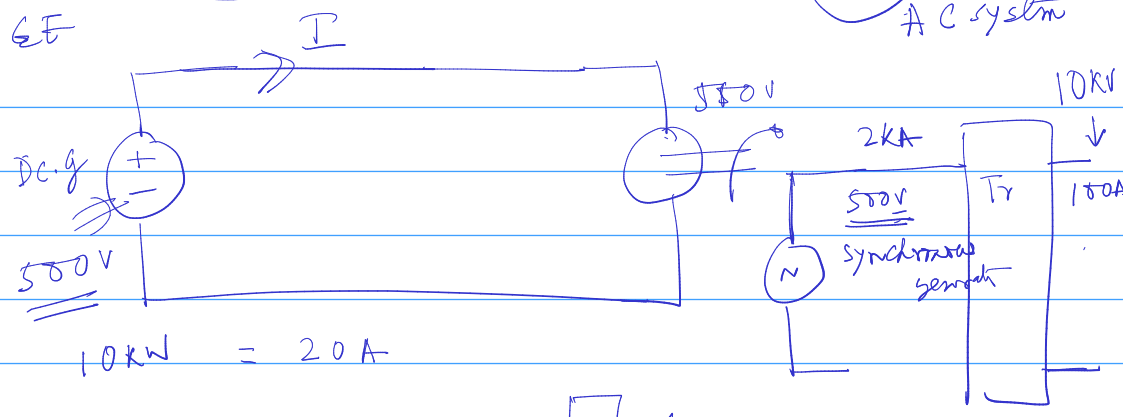
$L = N^2/R$

Transformer

Edisson – GE

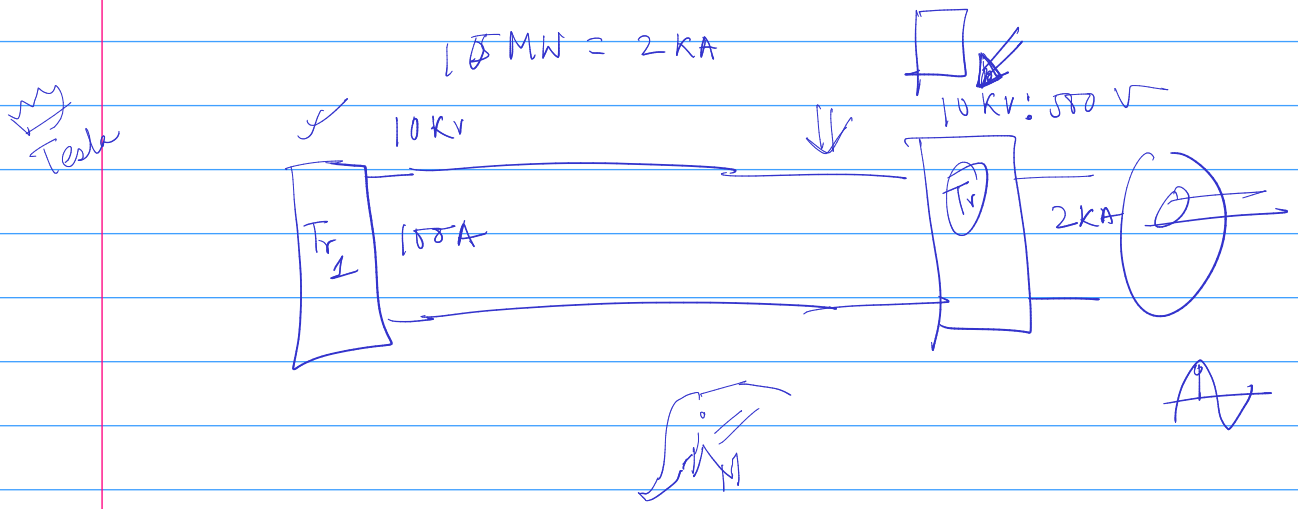
Tesla

A C system



10KW = 20A

10MW = 2KA

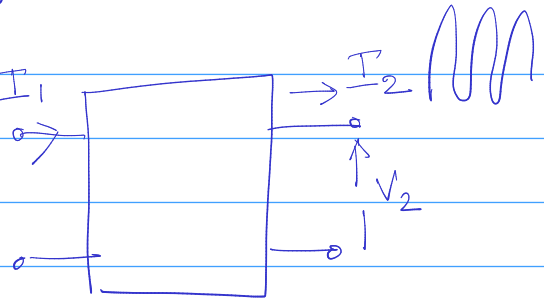


Class-4 Transformer 23/01/2026

— Single phase

Transformer works only for AC circuit

$100 \sin 2\pi 50t$



$1000 \sin 2\pi 50t$

$\frac{V_1}{V_2} = \left(\frac{N_1}{N_2}\right) \rightarrow (1)$

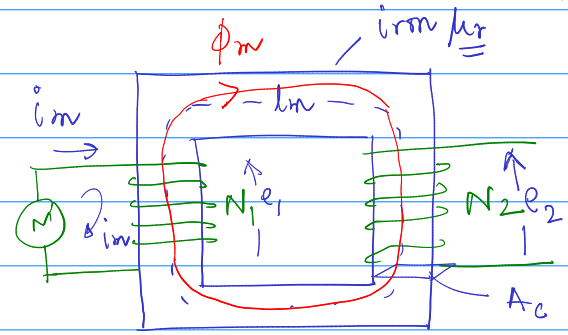
$\frac{I_1}{I_2} = \left(\frac{N_2}{N_1}\right) \rightarrow (2)$

$\frac{V_1 I_1}{V_2 I_2} = 1$ input power = Output power

$v_1 = V_m \sin \omega t$ $\quad \phi_1 = (V_1)$ $\quad V_1 = rms(v_1) = \left(\frac{V_m}{\sqrt{2}}\right)$

V_m = peak voltage of v_1

$\frac{N i_m}{R} = \phi_m$



$L_m = \frac{N_1^2}{R_m}$

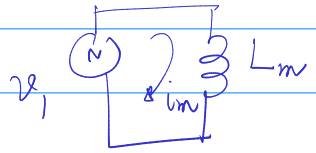
$R_m = \left(\frac{l_m}{\mu_0 \mu_r A_c}\right)$

circuit is linear (as it is unsaturated)

L_m = magnetizing inductance

$\omega = 2\pi \cdot 50 \cdot t$

$v_1 = V_m \cos \omega t$



$v_1 = L_m \frac{di_m}{dt}$

$i_m = \int \frac{V_m}{L_m} \cos \omega t \, dt = \left(\frac{V_m}{\omega L_m} \sin \omega t\right)$ $\quad \phi_m = |i_m| = \frac{V_m}{\omega L_m}$

$\phi_m = \frac{N V_m}{\omega L_m R} \sin \omega t = \phi \sin \omega t$ $\quad I_m = \frac{V_m}{(X_m)}$

$e_2 = -\frac{d}{dt}(N_2 \phi_m) = -\frac{d}{dt}(N_2 \phi \sin \omega t)$

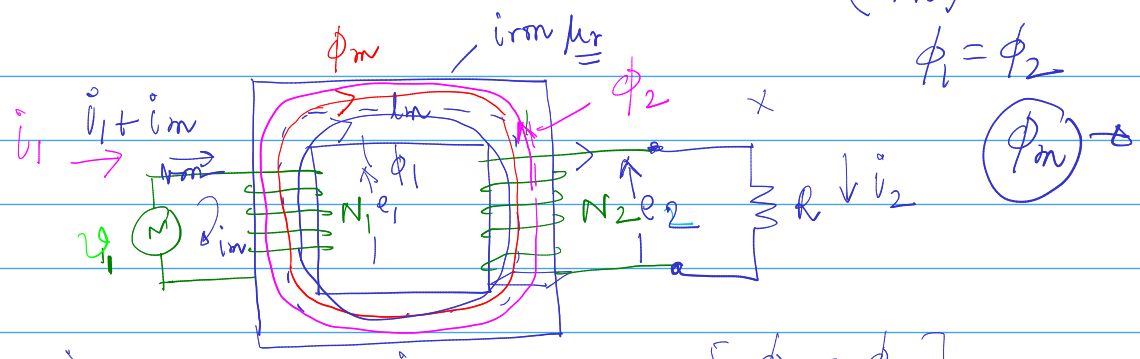
$= -N_2 \phi \omega \cos \omega t = N_2 \phi \omega \sin(\omega t - \pi/2)$

$e_1 = -\frac{d}{dt}(N_1 \phi_m) = N_1 \phi \omega \sin(\omega t - \pi/2)$

$\left(\frac{e_1}{e_2}\right) = \left(\frac{N_1}{N_2}\right)$

$i_2 = (e_2/R)$

$\phi_1 = \phi_2$

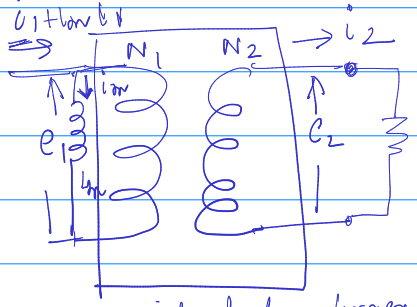


$\phi_2 = \frac{N_2 i_2}{\mathcal{R}}$ $\quad \phi_1 = \frac{N_1 i_1}{\mathcal{R}}$ $\quad [\phi_1 = \phi_2]$

$N_1 i_1 = N_2 i_2$ $\quad i\left(i_1/i_2 = \frac{N_2}{N_1}\right)$

$\frac{e_1}{e_2} = \frac{N_1}{N_2}$ $\quad i\quad \frac{i_1}{i_2} = \frac{N_2}{N_1} \rightarrow \frac{e_1 i_1}{e_2 i_2} = 1$ $\quad$ input power = output power

Ideal Tranformer $\rightarrow$ $\boxed{\frac{e_1}{e_2} = \frac{N_1}{N_2} = \frac{i_2}{i_1}}$



ideal transformer

$N_1 i_m = \phi \mathcal{R}$ $\quad \mathcal{R} = \frac{l_m}{\mu_0 \mu_r \cdot A_c}$ $\quad (i_m \rightarrow 0)$

$N_1 i_m = \frac{\phi l_m}{\mu_0 \mu_r A_c}$ $\quad \mu_r \rightarrow \infty$

$i_m \rightarrow 0$ $\quad \mu_r \rightarrow \infty$

Tranformer 200V/400V $\rightarrow$ 4 kW (power)

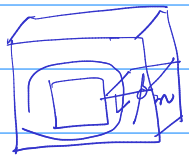
primary $\quad$ secondary side $\quad$ find I_1, I_2, N_2

$\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

$N_1 = 100$

$N_2 = 200$
$I_2 = 10\,A$
$I_1 = 20\,A$

$\phi_m = \hat{\phi}_m \cos \omega t$

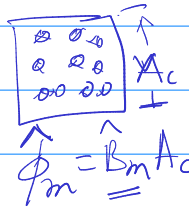


$e_1 = -\frac{d}{dt}(N_1 \phi_m)$

$= -N_1 \frac{d}{dt}(\hat{\phi}_m \cos \omega t)$

$e_1 = +N_1 \hat{\phi}_m \omega \sin \omega t$

$= N_1 \cdot 2\pi \cdot f \cdot \hat{B}_m A_c \cdot \sin \omega t$



$e_1 = \hat{e} \sin \omega t$

E_1 = r.m.s value of e_1

$\hat{e} = N_1 2\pi f \cdot \hat{B}_m A_c$

$= \frac{N_1 \cdot 2\pi}{\sqrt{2}} \cdot f \hat{B}_m A_c$

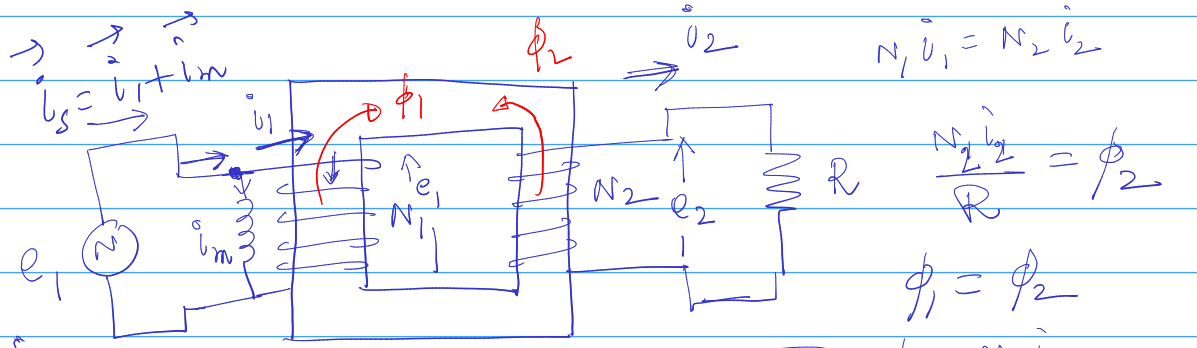
Class-5

27/01/2026

Modeling of Transformer

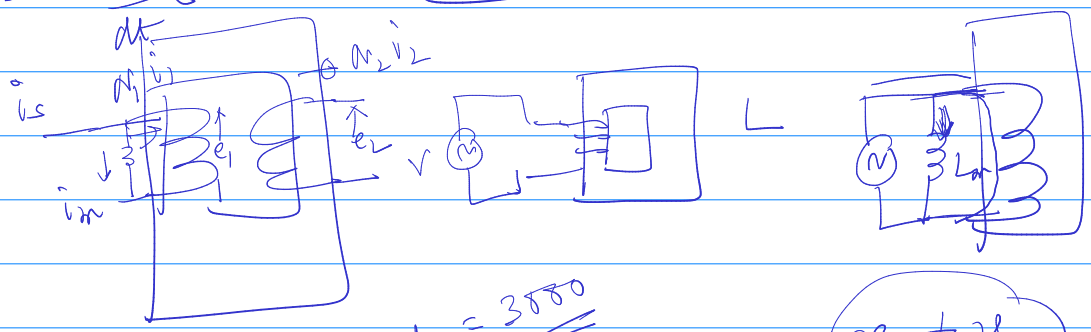
$$E_1 = 4.44 N_1 \cdot f \cdot \hat{B}_m A_c$$

[RMS value of induced primary side voltage



ϕ_m

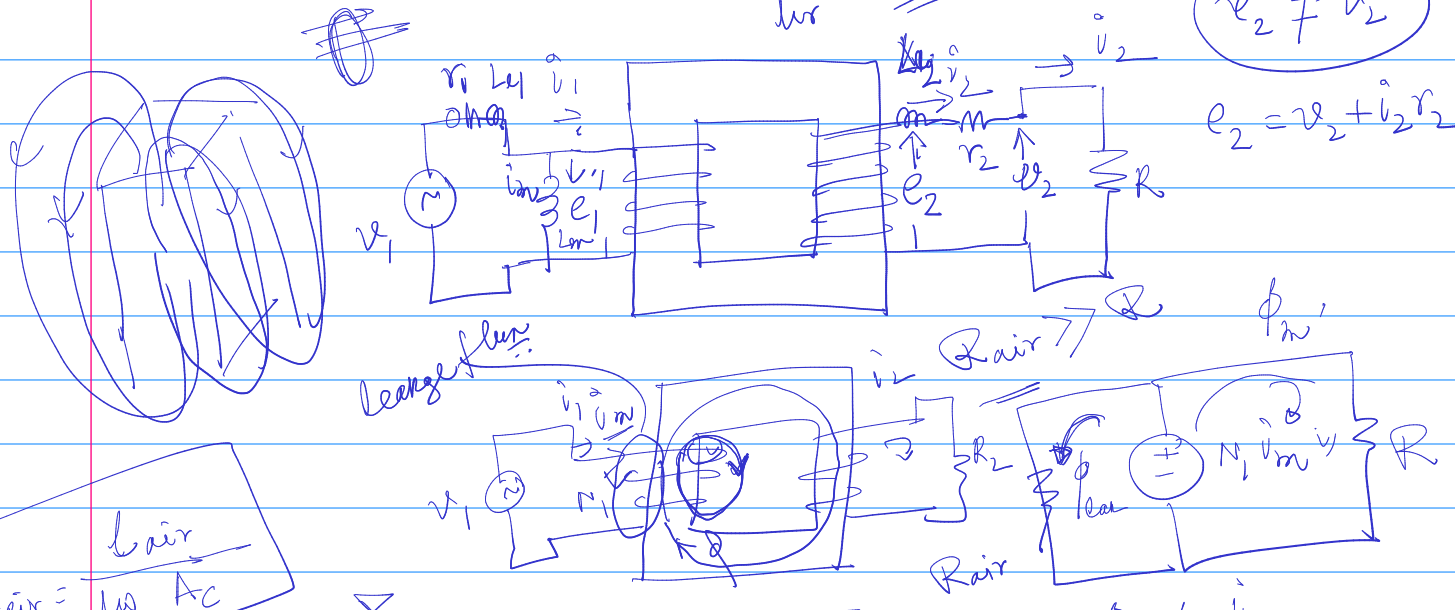
$\text{coil 1} = d(\phi_m + \phi_2 - \phi_1)$ $\quad N_1 i_1 = N_2 i_2$



$\mu_r = 3800$

$e_2 \neq v_2$

$e_2 = v_2 + i_2 r_2$



$$R_{air} = \frac{l_{air}}{\mu_0 A_c}$$

$\phi \propto i_m$

L_{ℓ_1} = Leakage inductance

X_{ℓ_1} = primary side leakage reactance

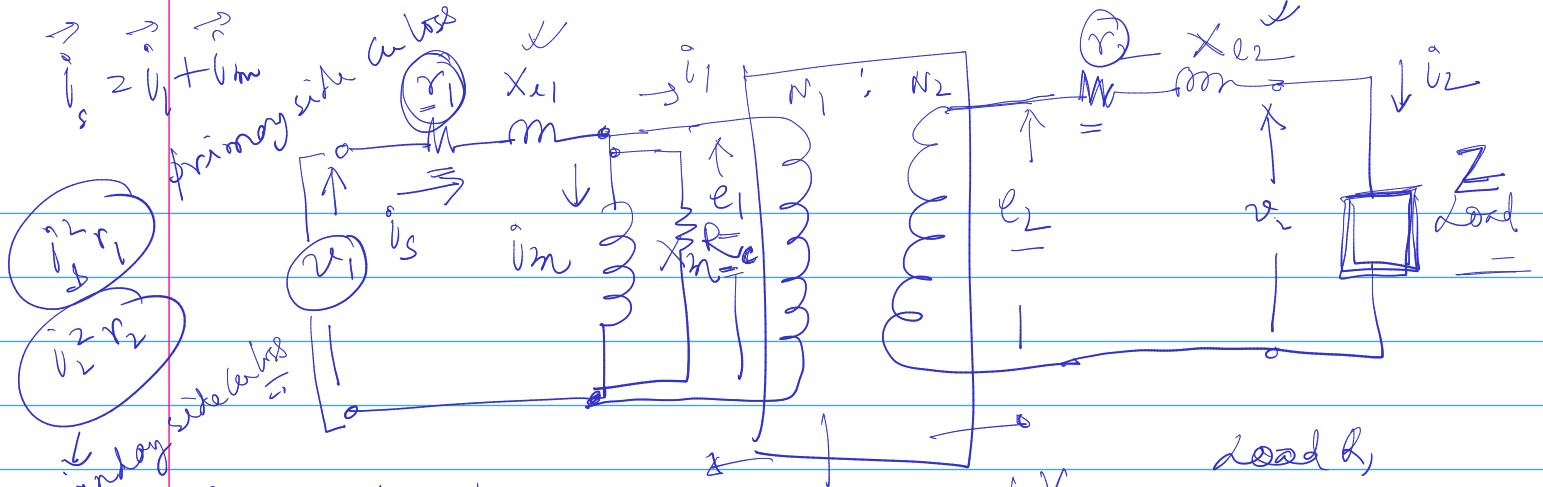
$X_{\ell_1} = 2\pi f L_{\ell_1}$

L_{ℓ_2} = Leakage inductance of coil 2

X_{ℓ_2} = Leakage reactance of coil 2

L_m = magnetizing inductance of the transformer

X_m = " reactance of the transformer



R, L load $\quad Z = \sqrt{R^2 + X_L^2} \quad i_2 = \left(\frac{V_2}{Z}\right)$

$X_L = (2\pi f L)$

load R,

R, L, R, L, C,

L, C

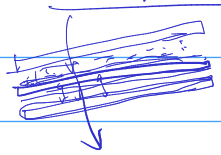
480 : 800 V

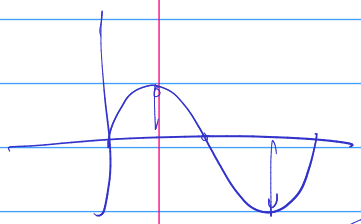
Core loss $= \left(\frac{V_x^2}{R}\right) \approx$ fixed loss (1)

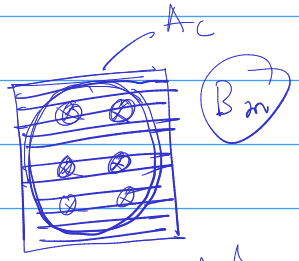
Class-1

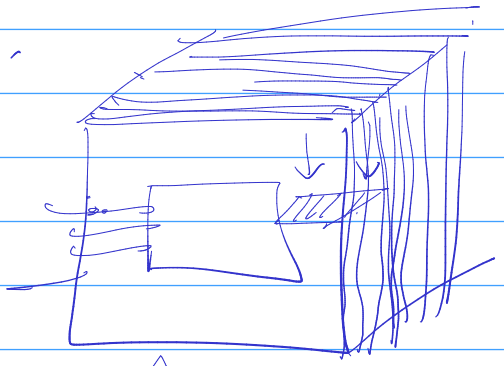
29/01/2026

Transformer core
- Eddy current loss
- Hysteresis loss.









$\phi = \phi_m \sin \omega t$

$V_{eddy} \propto e_1$

$e = -\frac{d\phi}{dt}$

$e = -\phi_m \omega \cos \omega t$

Eddy voltage

$I_{eddy} = \frac{\text{eddy voltage}}{\text{eddy resistance}}$

Eddy current loss $= I_{eddy}^2 R_{eddy} = \left(\frac{V_{eddy}^2}{R_{eddy}}\right) = \frac{V^2}{R}$

$P_{eddy} \propto \frac{1}{R_{eddy}} \uparrow \quad \left(\frac{V^2}{R_{eddy}} \downarrow\right)$

$e_1 = 4.44 f B_m A_c N_1$

$R_{eddy}^2 \frac{V}{I_{eddy}}$

$V_{eddy} \propto e_1$

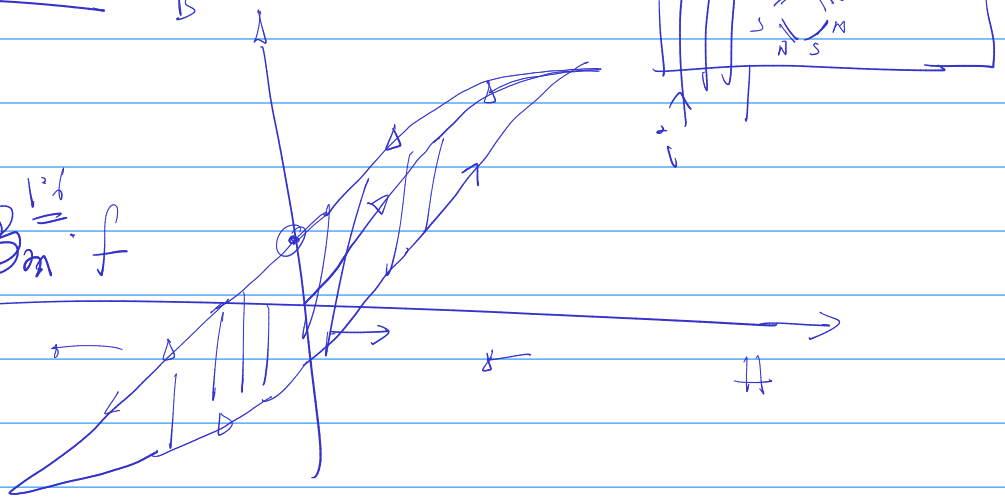
$e \propto (B_m f)$

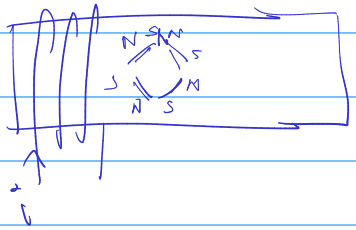
$P_{eddy} = \frac{V_{eddy}^2}{R}$

$P_{eddy} \propto \frac{e_1^2}{R}$

$\boxed{P_{eddy} \propto B_m^2 f^2}$

Hysteresis loss + eddy = Core loss





e

$P_{hysteresis} \propto B_m^{1.6} f$

Loss inside the transformer

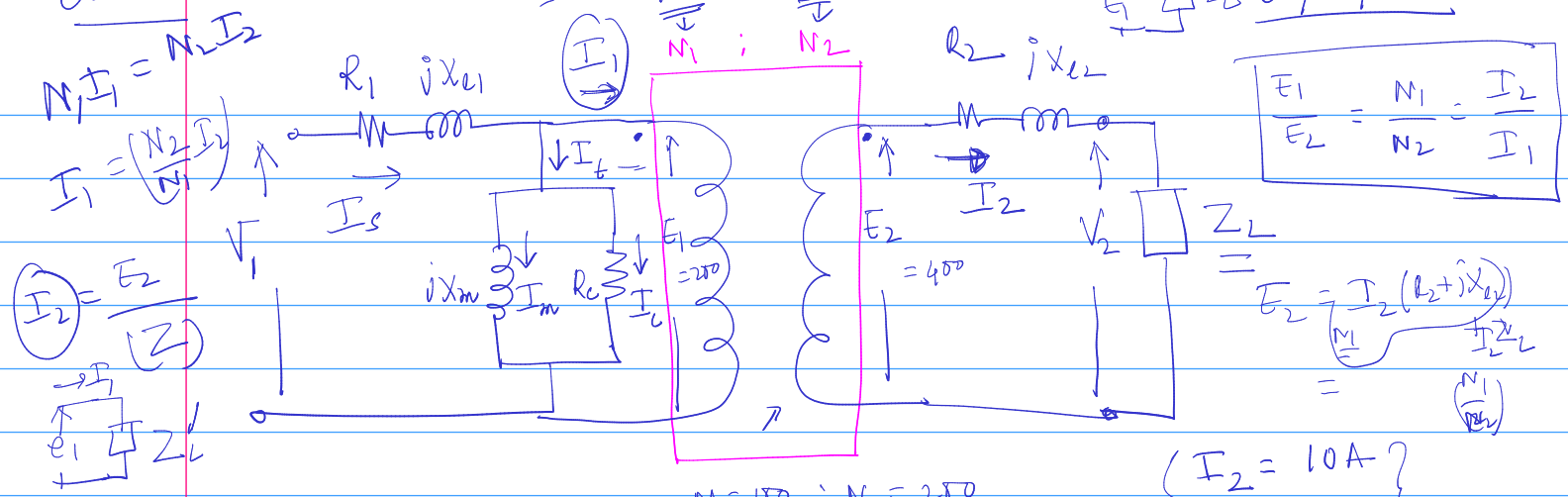
- fixed loss (core loss)
  - → Hysteresis + eddy current loss
  - $\frac{V^2}{R_e}$ (fictitious core loss resistance)
- variable loss (copper loss)
  - → copper conductor.
  - $(I_1^2 R_1 + I_2^2 R_2)$
  - Variable loss

Class 7

30/01/2026

$N_1 I_1 = N_2 I_2$

$I_1 = \left(\frac{N_2}{N_1}\right) I_2$



$\frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

$I_2 = \frac{E_2}{Z}$

$E_2 = I_2 (R_2 + jX_{L2}) + I_2 Z_L$

$= \left(\frac{N_1}{N_2}\right) E_1$

$N_1 = 100$; $N_2 = 250$

$E_1 = 250\,V$; $E_2 = 400\,V$ $\Rightarrow$ $I_2 = 10\,A$, $I_1 = 20\,A$

I_2' = secondary side current seen from the primary side

$N_1 I_1 = N_2 I_2$

$\left(N_1/N_2\right) I_1 = I_2$

$\frac{V_1}{V_2} = \frac{N_1}{N_2}$; $V_2 = \left(\frac{N_2}{N_1}\right) V_1$

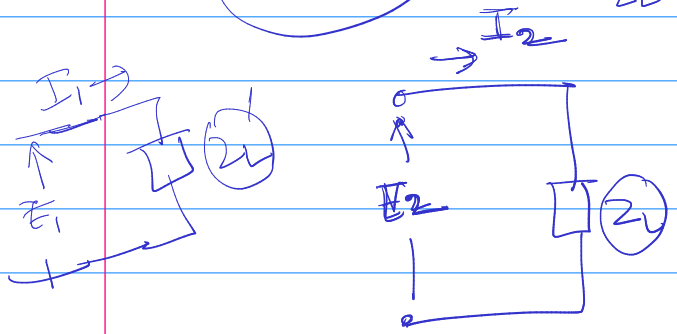
$I_2' = I_1$

$\left(\frac{N_1}{N_2}\right) I_1$

V_2' = Secondary side voltage seen from the primary side

$= (N_2/N_1) V_1$

$Z_L = (R_2 + jX_{L2} + Z)$



$\frac{E_2}{I_2} = (Z_L)$

$E_2 = \left(\frac{N_2}{N_1}\right) E_1$

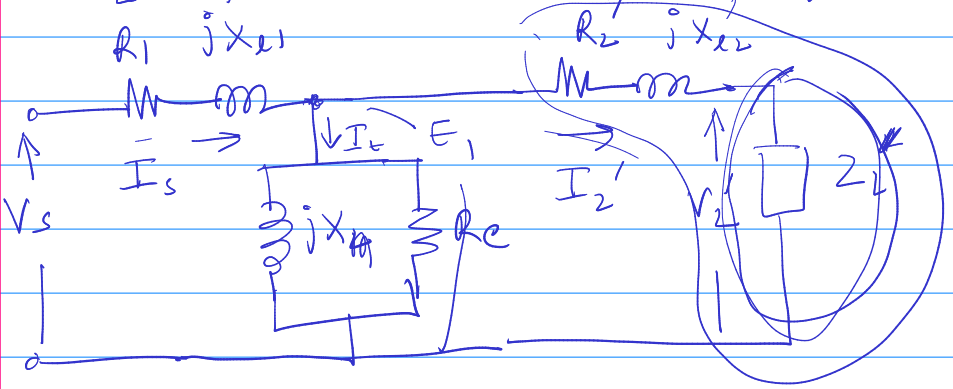
$\frac{\left(\frac{N_2}{N_1}\right) E_1}{\left(\frac{N_1}{N_2}\right) I_1} = Z_L$

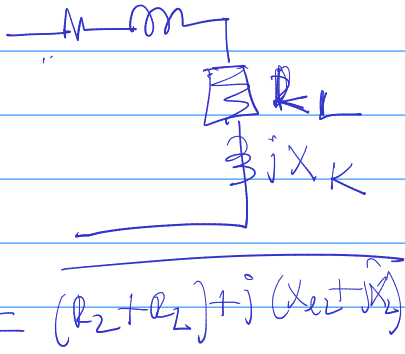
$\left(\frac{N_1}{N_2}\right)^2 Z_L$

$Z_L' \rightarrow$

$\Rightarrow V_1 / I_1 = \left(\frac{N_1}{N_2}\right)^2 Z_L$

Z_L referred to the primary side $\left(\frac{N_1}{N_2}\right)^2 Z_L = Z_L'$



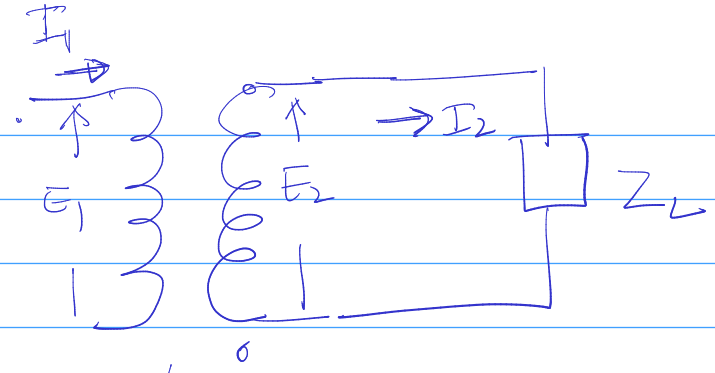


$= (R_2 + R_L) + j(X_{L2} + X_K)$

$= Z_L$

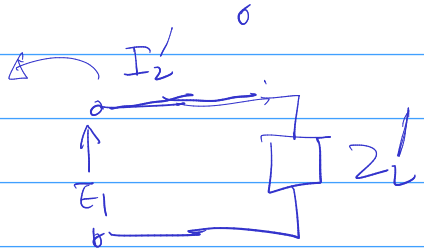
$N_1 I_1 = N_2 I_2$

$\frac{N_1}{N_2} I_1$



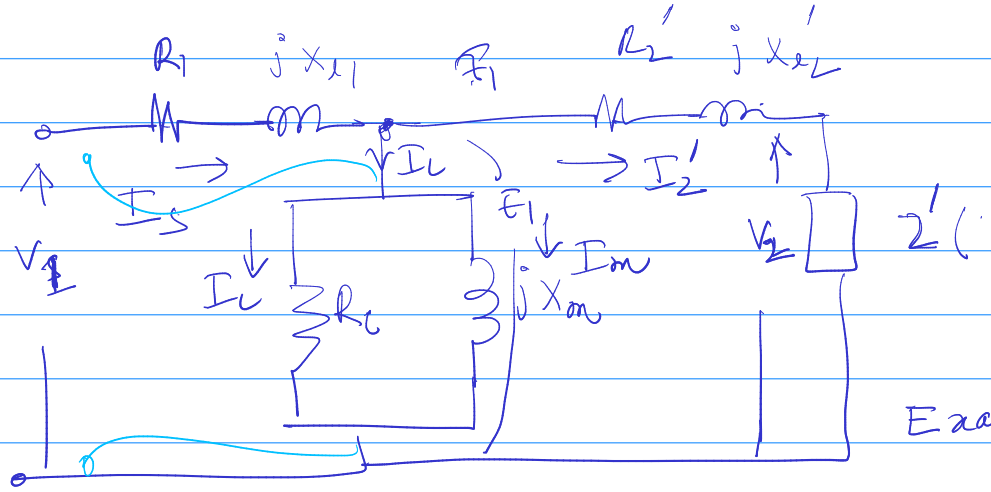
$R_L' = \left(\frac{N_1}{N_2}\right)^2 R_2$

$X_L' = \left(\frac{N_1}{N_2}\right)^2 X_L$

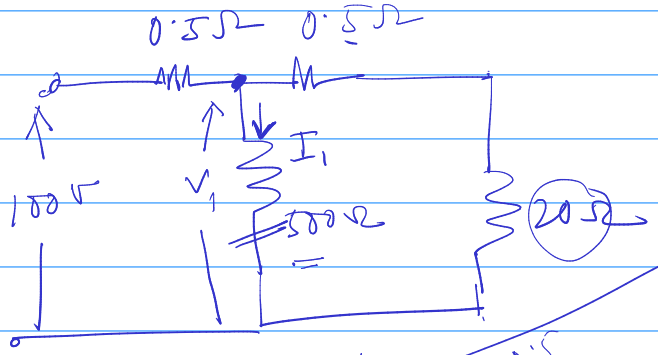


$\left(\frac{N_1}{N_2}\right)^2 Z_L$

$X_L' = \left(\frac{N_1}{N_2}\right)^2 X_L$



Exact equivalent ckt



$V_1 = 97.5$

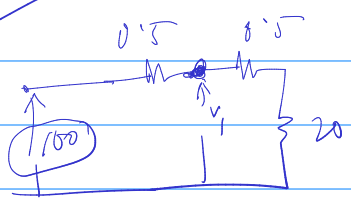
$I_1 = 0.195$

$V_1 \approx 100V$

$500\Omega \gg 20.5$

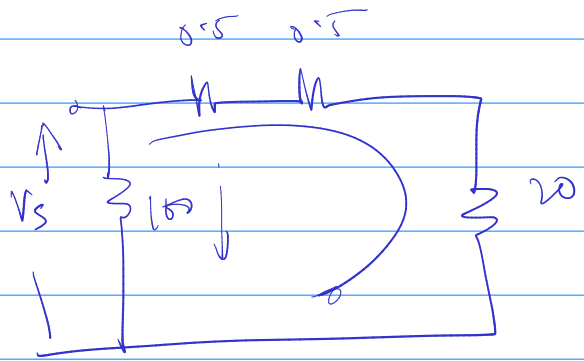
$I_1 = \frac{100}{500} = 0.2A$

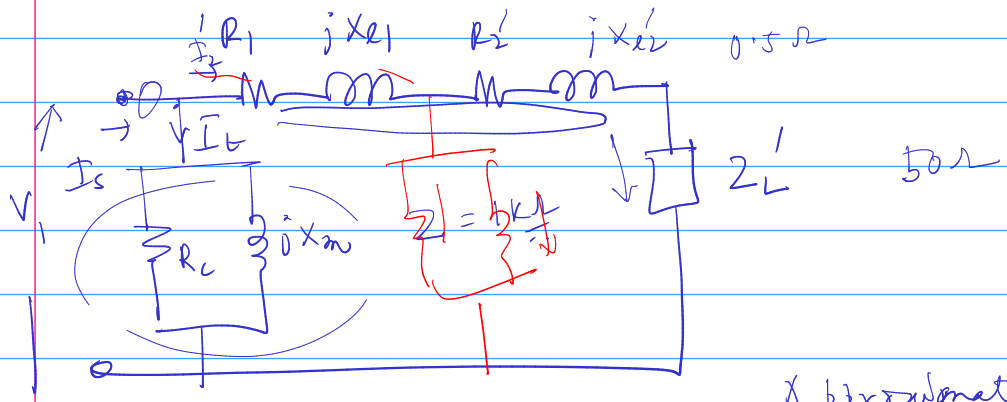
$20.5 \gg 0.5$



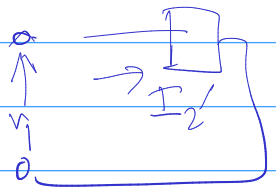
$\frac{100}{21} * 20.5$

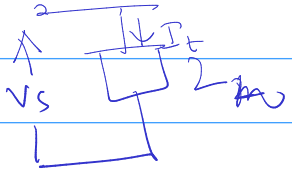
=





Approximate equivalent ckt



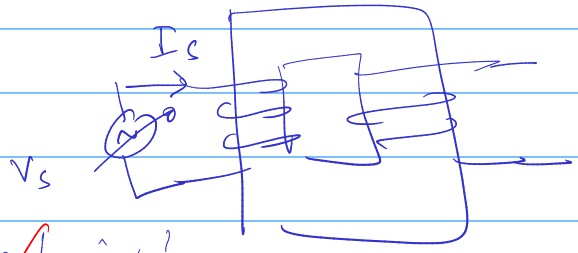


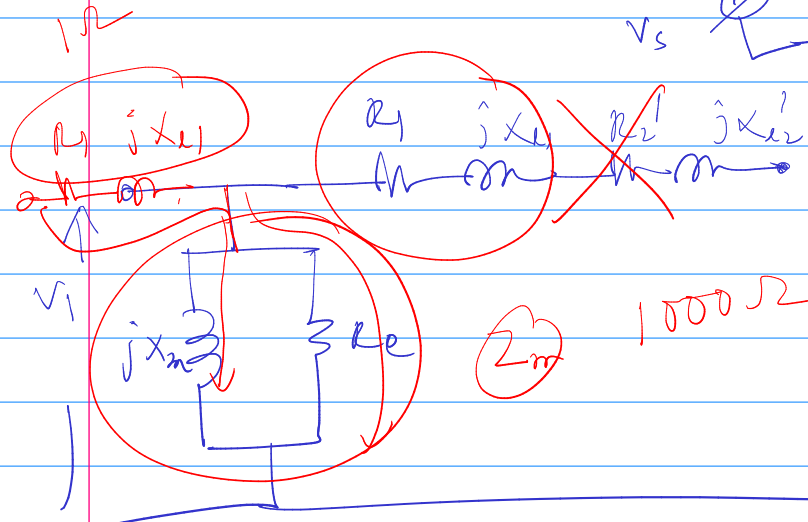
Open circuit test and short ckt test of the transformer

Find out parameters of the transformer

R_1, R_2 ; jX_{l1}, jX_{l2}', R_c, jX_m

Open ckt test of a transformer

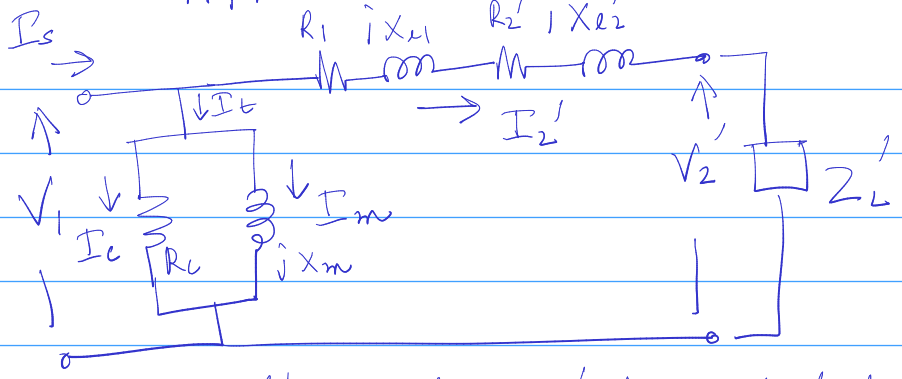




Class 8

02/02/2026

Approximate equivalent ckt =

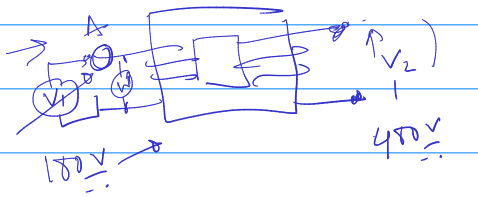


Open ckt test / short ckt test

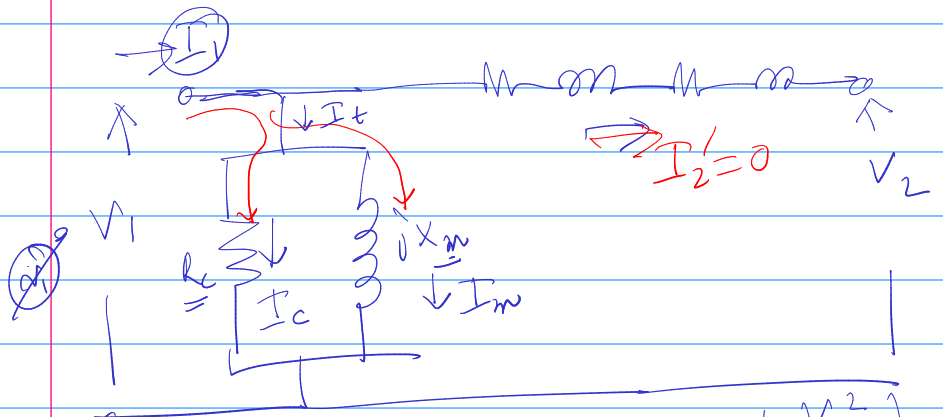
R_c, X_m / R_1, X_{L1}, R_2', X_{L2}'

Open ckt test

1) We keep the secondary side open
2) We apply rated voltage from the primary side
3) Measure the input voltage, input current and power consumption. 100V, 400V, 10KVA
4) We calculate R_c, X_m.



input Voltage (V), Input current (I), input power (W)



$\frac{I_1}{} = (rms)$

V_1, W_1

$V_1 = 100\,V$

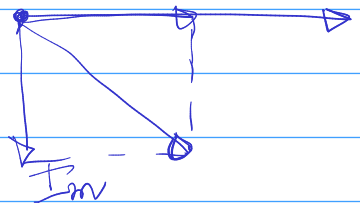
$W_1 = 100\,W$

$W_1 = \frac{V_1^2}{R_c}$ $\quad R_c = \left(\frac{V_1^2}{W_1}\right)$ $\quad R_c = \frac{(100)^2}{100} = 100\,\Omega$

$I_c = \frac{V_1}{R_c}$ $\quad I_m = \frac{V_1}{jX_m} = \frac{V_1 \angle 0}{X_m \angle 90} = \left(\frac{V_1}{X_m}\right) \angle -90$

$I_c = \frac{150}{150} = 1A$

$I_c = \frac{V_1}{R_c}$



$I_m = \left(\frac{V_1}{X_m}\right)$

$V_1 = 100V$

$R_c = 150\Omega$

I_t

$I_m = \sqrt{I_t^2 - I_c^2}$

$I_t = I_c - jI_m$

$|I_t| = \sqrt{I_c^2 + I_m^2}$

$I_t^2 = 2A;$

$I_m^2 = I_t^2 - I_c^2$

$I_m = \sqrt{4-1} = (\sqrt{3})$

$I_m = \sqrt{I_{t^2} - I_{c^2}}$

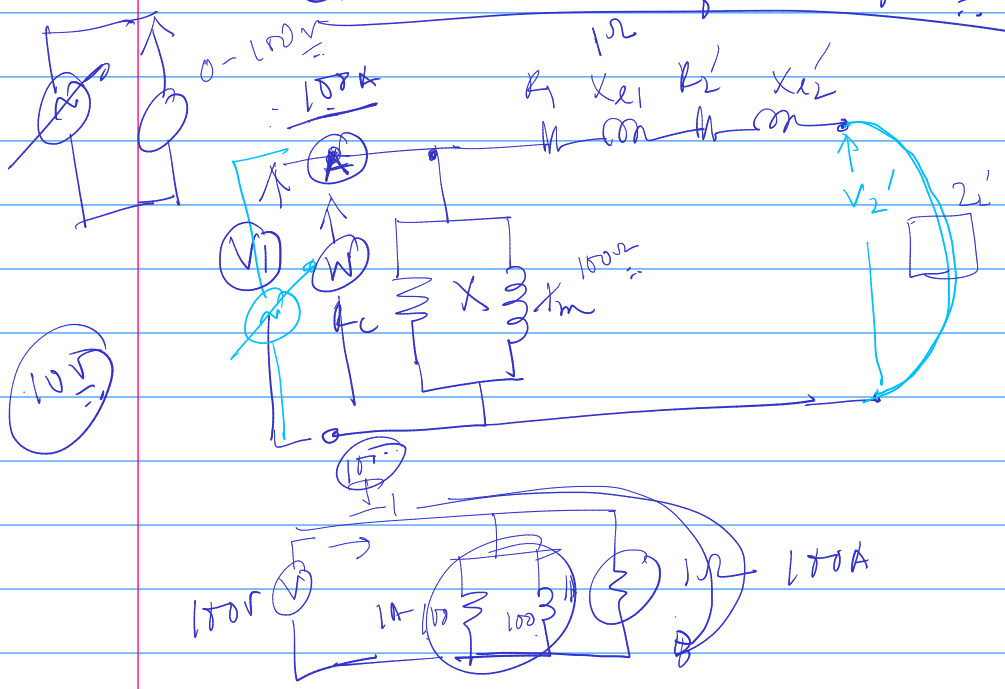
$X_m = \frac{V_1}{I_m} = \left(\frac{150}{\sqrt{3}}\right)$

$L_m = \frac{X_m}{2\pi f}$

$L_m = \left(\frac{150}{\sqrt{3} \cdot 2\pi \cdot 50}\right)$

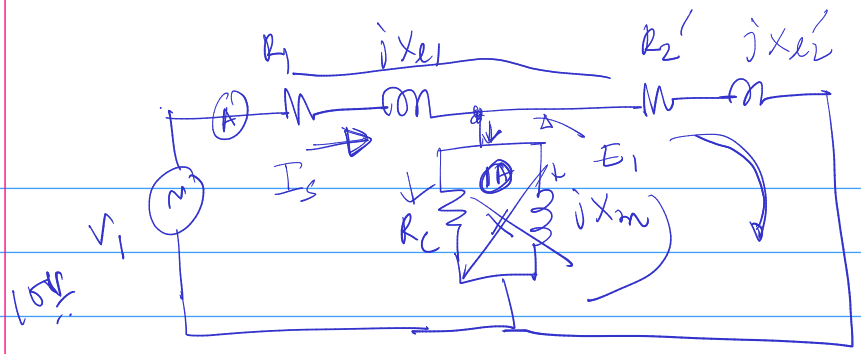
(1) $I = \frac{10KVA}{150} = 100A$ 150V : 450V / 10KVA

Short circuit test of a transformer (1Ω) $(R_1, R_2', X_{e1}, X_{e2}')$



1) short the secondary side

2) increase the input voltage slowly with an autotransformer. till rated current flows through the primary side.



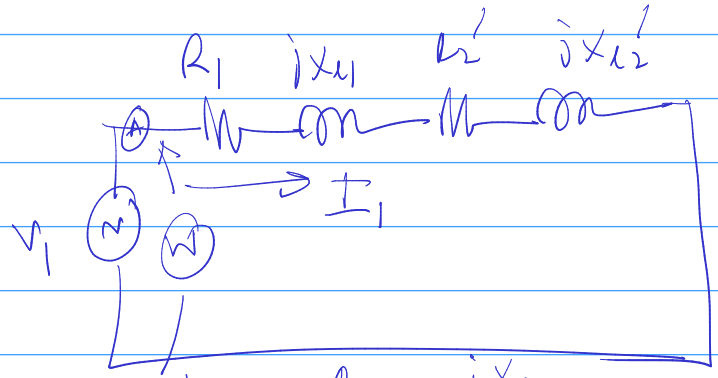
$$I_2' = \frac{E_1}{(R_2' + jX_2')}$$

$$I_2' = 100A, 0°A$$

$$I_s = I_2' + I_t$$

$$= 100A + 0.1$$

$$\approx I_2'$$

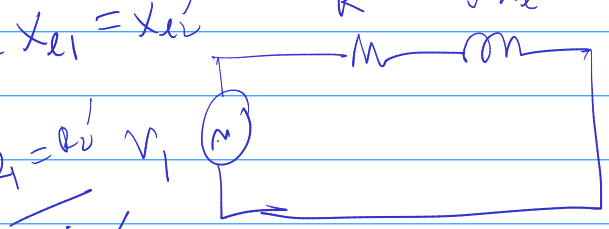


$$R_1 \approx R_2' \qquad X_{L1} \approx X_{L2}'$$

$$R = R_1 + R_2'$$

$$X_L = X_{L1} + X_{L2}'$$

$$Z = (R + jX_L) = \sqrt{R^2 + X_L^2}$$



$$X_L/2 = X_{L1} = X_{L2}'$$

$$R/2 = R_1 = R_2'$$

$$I_1^2 R = W \qquad R = \left(\frac{W}{I_1^2}\right)$$

$$Z = \left(\frac{V_1}{I_1}\right) = |Z|$$

$$X_L = \sqrt{Z^2 - R^2}$$

100 KVA

11KV / 400V

Distribution transformer

$$11KV/\sqrt{3} : 400/\sqrt{3}$$

$$6350 : 230$$

10-20

$$P = \sqrt{3} V_L I_L$$

$$I_L = \frac{100 \times 1000}{\sqrt{3} \cdot 11 \cdot KV} = \frac{100}{\sqrt{3} \cdot 11}$$

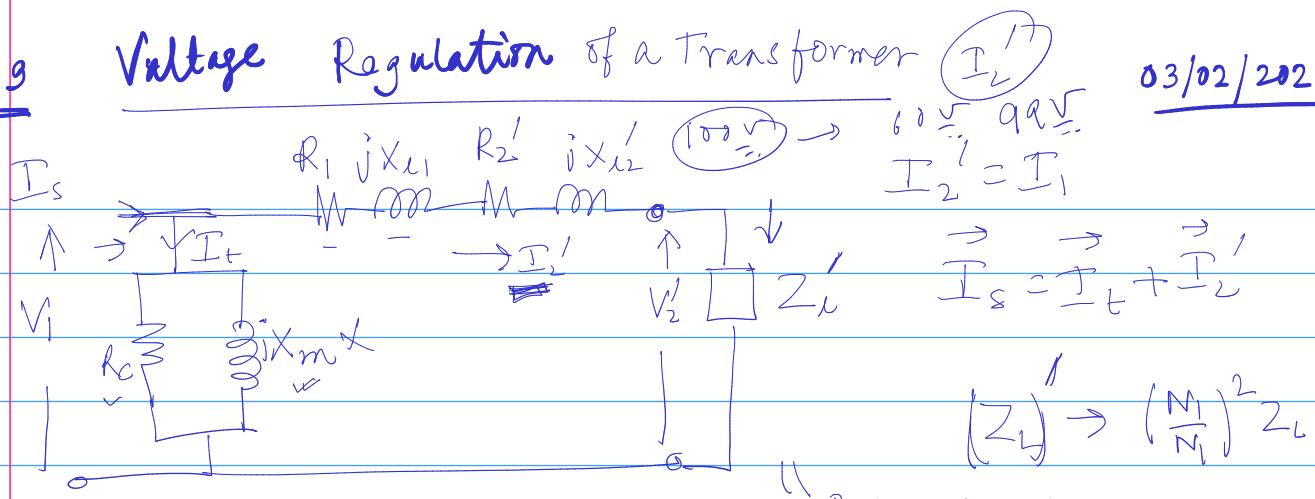
$$5A : 138A$$

$$I_s = \frac{6350}{230} * 5 = 138$$

(*) Prefer to conduct open ckt test from the LV side keeping HV side open.

(*) Preferred to conduct short ckt test from the HV side keeping LV side shorted.

class-9 **Voltage Regulation** of a Transformer 03/02/2026



$\vec{I}_s = \vec{I}_t + \vec{I}_2'$

$I_2' = I_1$

$(Z_L)' \rightarrow \left(\frac{N_1}{N_2}\right)^2 Z_L$

Behave ideal transformer

400 V : 100 V , 10 KVA

load at no load $\Rightarrow$ 400 V : 100 V

full load 10 KVA $\Rightarrow$ 400 V : 100 V

100 A

Voltage regulation = $\frac{\text{No load voltage} - \text{full load voltage}}{\text{No load voltage}} \times 100\%$

No load = 100 V

full load = 90 V

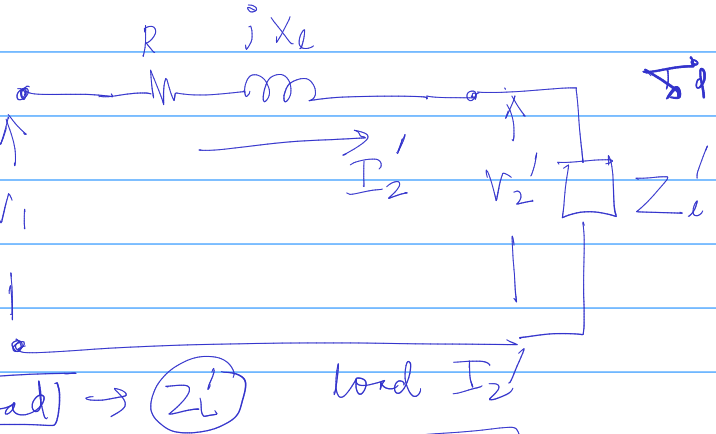
V.R = $\frac{100-90}{100} \times 100 = 10\%$

No load voltage = 100 V

full load voltage = 110 V

V.R = $\frac{100-110}{100} \times 110 = -10\%$

$R = R_1 + R_2'$; $X_e = X_{L1} + X_{L2}'$



At no load $V_2' = V_1$

Load $\rightarrow$ Z_L'

Z_L' = 1) Resistive load
2) Inductive load (R, L)
3) Capacitive load (R, C)

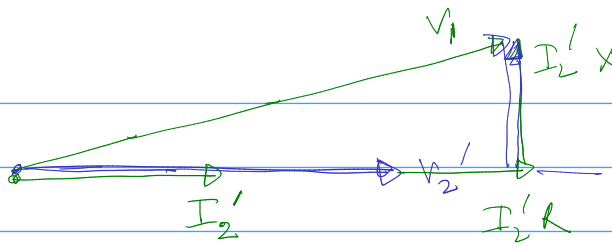
Resistive load $\Rightarrow$ I_2' ; V_2 phase

phase relationship b/w V_2' & I_2' is known as load power factor.

(Unity power factor)

$V_1 = V_2' + I_2' R + j I_2' X_L$

$V_1 = \sqrt{(V_2' + I_2' R)^2 + (I_2' X_L)^2}$



$I_2' X_L \ll V_2'$

$V_2' = 100\,V$

$I_2' X_L = 1\,V$

$I_2' R = 10\,V$

$\left(\frac{V_1 - V_2'}{V_1}\right) = V.R.$

$V_1 = \sqrt{(100+1)^2 + (1)^2} = 101$

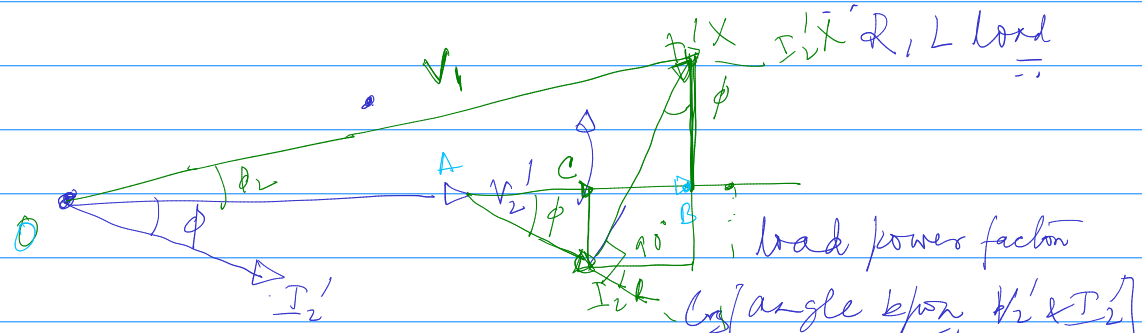
$V_1 \approx V_2' + I_2' R$

$V.R. = \frac{V_2' + I_2' R - V_2'}{V_1}$

$= \left(\frac{I_2' R}{V_1}\right) \times 100\%$

Inductive load

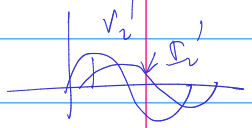
$OB \approx V_1$



lead power factor

Cos[angle b/w V_2' & I_2']

$= \cos\phi$ lags



$OB = OA + AC + BC$

$= V_2' + I_2' R \cos\phi + I_2' X \sin\phi$

$V_1 = V_2' + I_2' R \cos\phi + I_2' X \sin\phi$

$V.R. = \frac{V_2' + I_2' R \cos\phi + I_2' X \sin\phi - V_2'}{V_1}$

$\boxed{V.R} = \left(\frac{I_2' R \cos\phi + I_2' X \sin\phi}{V_1}\right) \times 100\%$

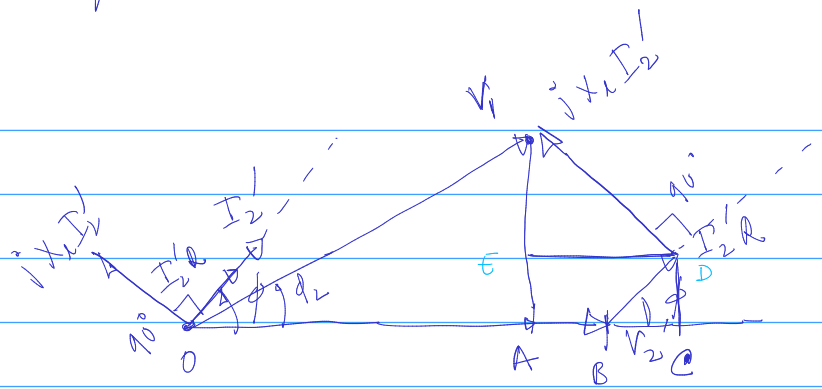
$V_1 = 100\,V$; $I_2' = 10\,A$; $R = 0.1\,\Omega$; $X = 0.1\,\Omega$

$\cos\phi = 0.7$.

find $V.R.$: $= \frac{10 \cdot 0.1 \cdot 0.7 + 10 \times 0.1 \times 0.7}{100} \times 100 =$ 14%

$V_1 = V_2' + I_2' R + j I_2' X_2 \quad I_2' j X_2$

Capacitive load



$\vec{V_1} = OA$

$V_1 = OA = OB + BC - AC = V_2' + I_2' R \cos\phi - I_2' X \sin\phi$

$V.R = \dfrac{V_2' + I_2' R \cos\phi - I_2' X \sin\phi - V_2'}{V_1} \times 100\%$

$V.R = \dfrac{I_2' R \cos\phi - I_2' X \sin\phi}{V_1} \times 100\%$

V.R

Zero voltage regulation $= \dfrac{0}{V_1} \times 100\%$

$I_2' R \cos\phi = I_2' X \sin\phi$

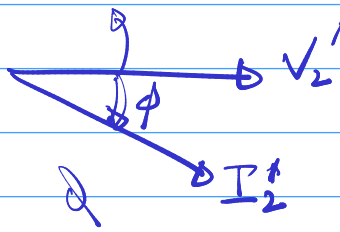
$\tan\phi = R/X$

Maximum voltage regulation $\dfrac{d(V.R)}{d\phi} = 0$

Class-10 **Voltage regulation of a transformer** 05/02/2024

V.R. (Inductive load) = $\frac{I_2' R\cos\phi + I_2' X \sin\phi}{V_1}$ ✓✓

V.R. (Capacitive load) = $\frac{I_2' R\cos\phi - I_2' X \sin\phi}{V_1}$



ϕ = +ve

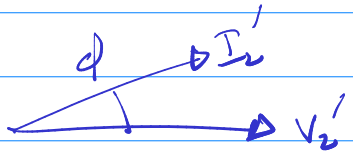
$\cos\phi$ +ve

$\sin\phi$ +ve

$I_2' R\cos(-\phi) + I_2' X \sin(-\phi)$

$= I_2' R\cos\phi - I_2' X \sin\phi$

Capacitive load



ϕ = -ve

Zero voltage regulation V.R = 0

$I_2' R\cos\phi - I_2' X \sin\phi = 0$

$X \sin\phi = R\cos\phi$

$\tan\phi = R/X$

$\phi = \tan^{-1}(R/X)$

maximum regulation:

$\frac{d}{d\phi}(V.R) = 0$

V.R. = $I_2' R\cos\phi + I_2' X \sin\phi$

$\frac{d}{d\phi}(VR) = -I_2' R\sin\phi + I_2' X\cos\phi = 0$

$I_2' R\sin\phi = I_2' X\cos\phi$

$\tan\phi = X/R$

20%

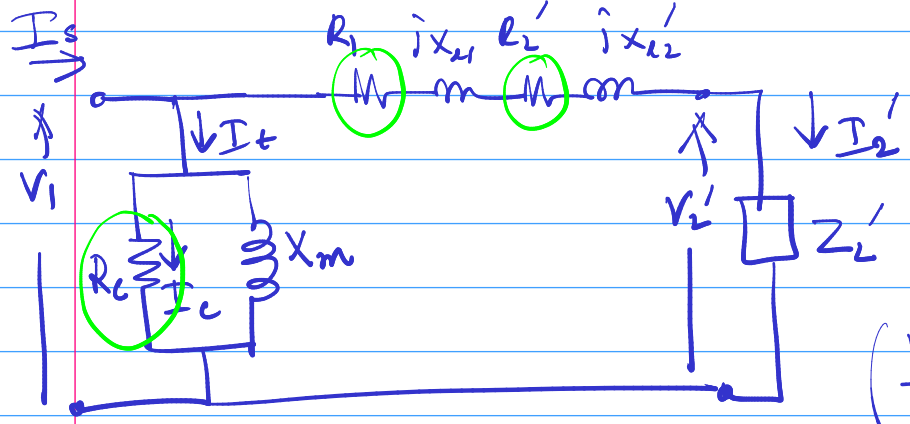
Max ± 5%

Performance

- Voltage Regulation
- efficiency

89%

98%



$\cos\phi$

$[I_c^2 R_c]$ = core loss

$\left(\frac{V_1^2}{R_c}\right)$

eddy current loss, Hysteresis loss

Core loss = iron loss = fixed loss

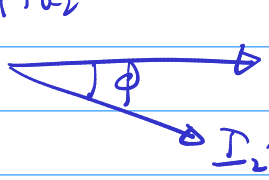
$I_2'^2(R_1+R_2')$ = Copper loss of a transformer $\propto I_2'^2$

= Variable loss.

efficiency

$$\eta\% = \frac{\text{Output power}}{\text{Input power}} \times 100\%$$

$R = R_1 + R_2'$



V_2'

Output power = $V_2' I_2' \cos\phi$

Input power = $V_2' I_2' \cos\phi + P_{iron} + P_{cu}$

$= V_2' I_2' \cos\phi + P_{iron} + I_2'^2 R$

4 to
4 05
3 95

$$\eta\% = \frac{V_2' I_2' \cos\phi}{V_2' I_2' \cos\phi + P_{iron} + P_{cu}}$$

$P_{cu} = I_2'^2 R$

at a fixed p.f.

$$\eta\% = \frac{V_2' \cos\phi}{V_2' \cos\phi + \frac{P_{iron}}{I_2'} + \frac{P_{cu}}{I_2'}} = \frac{V_2' \cos\phi}{V_2' \cos\phi + \frac{P_{iron}}{I_2'} + I_2' R}$$

$$D = V_2' \cos\phi + \frac{P_{iron}}{I_2'} + I_2' R$$

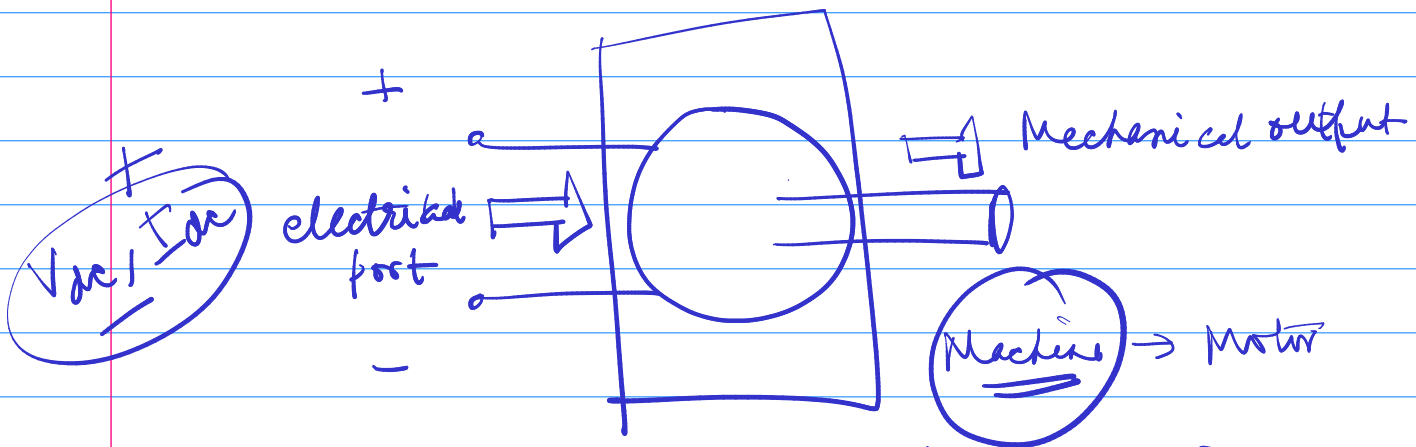
$$\frac{dD}{dI_2'} = -\frac{P_{iron}}{I_2'^2} + R = 0$$

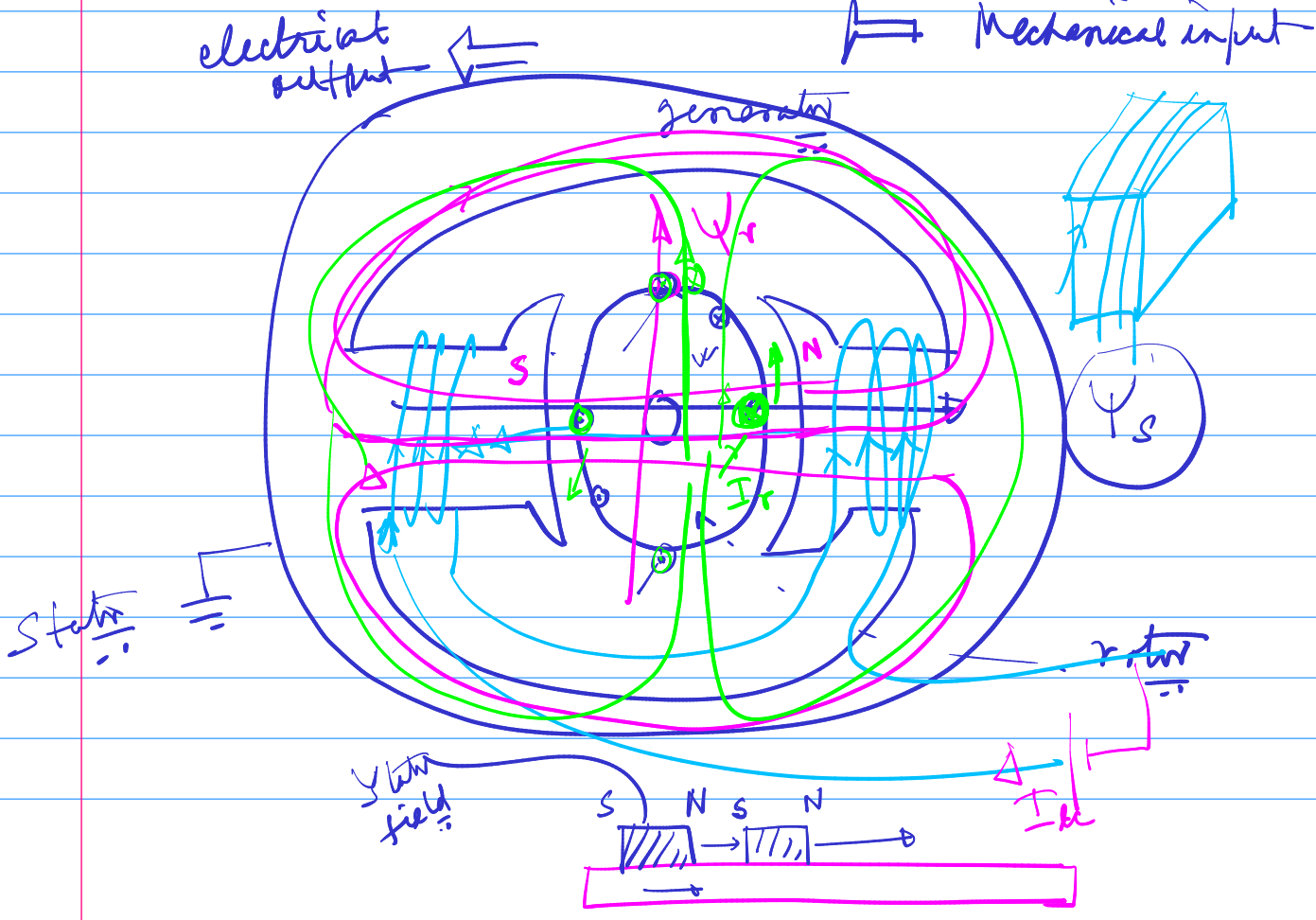
$$P_{iron} = I_2'^2 R$$

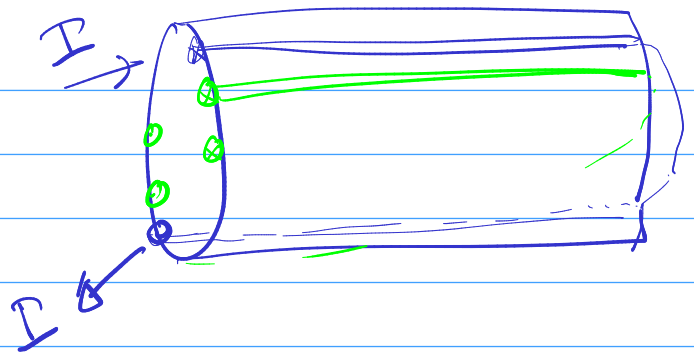
Maximum efficiency $\Rightarrow$ $P_{iron} = P_{cu}$

→ 3-ϕ Transformer → 11 operation

D.C. Motor / D.C Generator






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